**A**　【基本】次の式を展開しなさい。

(1) $(x+4)(x+9) = x^2+13x+36$

(2) $(x+8)^2 = x^2+16x+64$

(3) $3x(2x+1) = 6x^2+3x$

(4) $(a+8)(a-8) = a^2-64$

(5) $(x-3)(x-8) = x^2-11x+24$

(6) $(y-9)^2 = y^2-18y+81$

(7) $7a(a-2) = 7a^2-14a$

(8) $(b+7)(b-7) = b^2-49$

(9) $(x+2)(x-11) = x^2-9x-22$

(10) $(m+4)^2 = m^2+8m+16$

(11) $-x(x+8) = -x^2-8x$

(12) $(n+5)(n-5) = n^2-25$

(13) $(a-4)(a+5) = a^2+a-20$

(14) $(x-2)^2 = x^2-4x+4$

(15) $(x+3)(x-10) = x^2-7x-30$

(16) $(x+2)(x+12) = x^2+14x+24$

## A　【基本】次の式を因数分解しなさい。

(1) $x^2+10x+24 = (x+4)(x+6)$

(2) $8x+24 = 8(x+3)$

(3) $a^2-4 = (a+2)(a-2)$

(4) $b^2+6b+9 = (b+3)^2$

(5) $x^2-12x+35 = (x-5)(x-7)$

(6) $x^2-7x = x(x-7)$

(7) $y^2-121 = (y+11)(y-11)$

(8) $n^2-8n+16 = (n-4)^2$

(9) $x^2+5x-24 = (x+8)(x-3)$

(10) $5a-15b = 5(a-3b)$

(11) $m^2-100 = (m+10)(m-10)$

(12) $x^2+20x+100 = (x+10)^2$

(13) $x^2-x-30 = (x+5)(x-6)$

(14) $a^2+12a+35 = (a+5)(a+7)$

(15) $64-x^2 = (8+x)(8-x)$

(16) $y^2-16y+63 = (y-7)(y-9)$

## B　【基本】次の式を因数分解しなさい。

(1) $x^2+14x+24 = (x+2)(x+12)$

(2) $4a^2+10a = 2a(2a+5)$

(3) $b^2-49 = (b+7)(b-7)$

(4) $a^2-18a+81 = (a-9)^2$

(5) $x^2-2x-35 = (x+5)(x-7)$

(6) $3xy-9y = 3y(x-3)$

(7) $y^2-4 = (y+2)(y-2)$

(8) $m^2+12m+36 = (m+6)^2$

(9) $x^2+15x+56 = (x+7)(x+8)$

(10) $12a-8b = 4(3a-2b)$

(11) $n^2-64 = (n+8)(n-8)$

(12) $y^2-14y+49 = (y-7)^2$

(13) $x^2+4x-21 = (x+7)(x-3)$

(14) $a^2-11a+28 = (a-4)(a-7)$

(15) $81-x^2 = (9+x)(9-x)$

(16) $x^2+9x-22 = (x+11)(x-2)$

**A**　【基本】(3)(4)(11)(12)は分母を有理化し、(7)は根号1つの形にしなさい。それ以外は計算しなさい。

(1) $\sqrt{24} = 2\sqrt{6}$

(2) $\sqrt{3} \times \sqrt{6} = 3\sqrt{2}$

(3) $\dfrac{1}{\sqrt{7}} = \dfrac{\sqrt{7}}{7}$

(4) $\dfrac{3}{\sqrt{2}} = \dfrac{3\sqrt{2}}{2}$

(5) $2\sqrt{5} + 4\sqrt{5} = 6\sqrt{5}$

(6) $\sqrt{45} \div \sqrt{5} = 3$

(7) $2\sqrt{2} = \sqrt{8}$

(8) $\sqrt{162} = 9\sqrt{2}$

(9) $\sqrt{48} - \sqrt{27} = \sqrt{3}$

(10) $2\sqrt{2} \times 4\sqrt{3} = 8\sqrt{6}$

(11) $\dfrac{6}{\sqrt{3}} = 2\sqrt{3}$

(12) $\dfrac{\sqrt{5}}{\sqrt{2}} = \dfrac{\sqrt{10}}{2}$

(13) $\sqrt{40} = 2\sqrt{10}$

(14) $8\sqrt{14} \div 2\sqrt{7} = 4\sqrt{2}$

(15) $\sqrt{18} + \sqrt{32} = 7\sqrt{2}$

(16) $\sqrt{20} - \sqrt{45} + \sqrt{80} = 3\sqrt{5}$